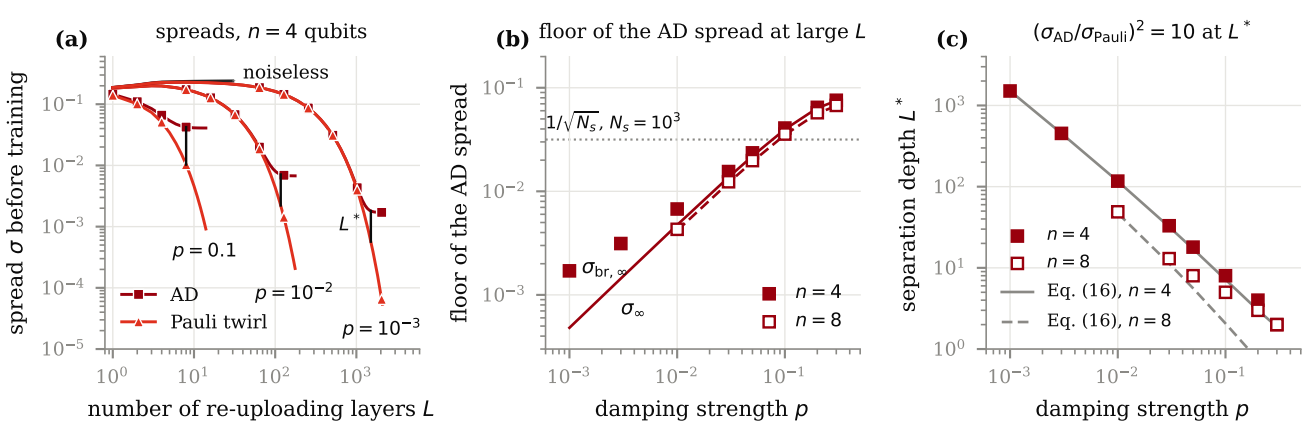
(a)
spreads, $n = 4$ qubits
noiseless
$p = 0.1$
$p = 10^{-2}$
$p = 10^{-3}$
L^*
AD
Pauli twirl
spread σ before training
number of re-uploading layers L
(b)
floor of the AD spread at large L
$1/\sqrt{N_s}$, $N_s = 10^3$
$\sigma_{\mathrm{br},\infty}$
σ_∞
$n = 4$
$n = 8$
floor of the AD spread
damping strength p
(c)
$(\sigma_{\mathrm{AD}}/\sigma_{\mathrm{Pauli}})^2 = 10$ at L^*
$n = 4$
$n = 8$
Eq. (16), $n = 4$
Eq. (16), $n = 8$
separation depth L^*
damping strength p